

Footprinting with Kali Tools

NetworkWalks — Week 2 Cybersecurity Project

1. Introduction

As part of my Week 2 Cybersecurity training with **NetworkWalks**, I performed a practical footprinting exercise using several reconnaissance tools available in Kali Linux.

The purpose of the exercise was to understand how publicly available information about a target can be collected and analyzed during the reconnaissance phase of a security assessment.

The exercise covered domain registration information, web technologies, DNS information, HTTP response headers, Web Application Firewall detection, and DNS record enumeration.

2. Objectives

The objectives of this exercise were to:

- Understand the reconnaissance and footprinting phase of a security assessment.
- Gather publicly available information about authorized targets.
- Identify technologies used by a target web application.
- Resolve domains to their associated IP addresses.
- Examine HTTP response headers.
- Identify potential Web Application Firewall (WAF) technologies.
- Enumerate publicly accessible DNS records.
- Document reconnaissance findings in a structured manner.

3. Tools Used

Tool	Purpose
WHOIS	Domain registration and ownership information
WhatWeb	Web technology identification
NSLookup	DNS and domain-to-IP resolution
cURL	HTTP response/header analysis
WAFW00F	WAF detection
DNSRecon	DNS record enumeration
Kali Linux	Security testing environment

4. Target and Authorization

The reconnaissance activities were performed against targets for which I had obtained permission to conduct security reconnaissance.

The testing was conducted within the scope authorized for the exercise and was intended for cybersecurity training and learning purposes.

Target: networkwalks.com

Authorization: Permission was obtained from the appropriate target/network owner before reconnaissance activities were conducted.

5. Footprinting Activities

5.1 WHOIS Enumeration

WHOIS was used to obtain publicly available domain registration information.

Command

```
whois networkwalks.com
```

Information examined

- Registrar
- Domain creation date
- Domain expiration date
- Domain status
- Name servers
- Public registration information

Result

```
kali@kali: ~/Documents/networkwalks-week2
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2023-11-12T10:08:42Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2585
Domain Status: clientDeleteProhibited https://icann.org/epp/clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp/clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp/clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-17T07:18:56Z +00

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in Verisign Global Registry
Services' ("Verisign") Whois database is provided by Verisign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. Verisign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this data only
for lawful purposes and that under no circumstances will you use this data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to Verisign (or its computer systems). The compilation,
repackaging, dissemination or other use of this data is expressly
prohibited without the prior written consent of Verisign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. Verisign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. Verisign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. Verisign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
```

Observation

The WHOIS investigation provided information about the target's domain registration and associated infrastructure.

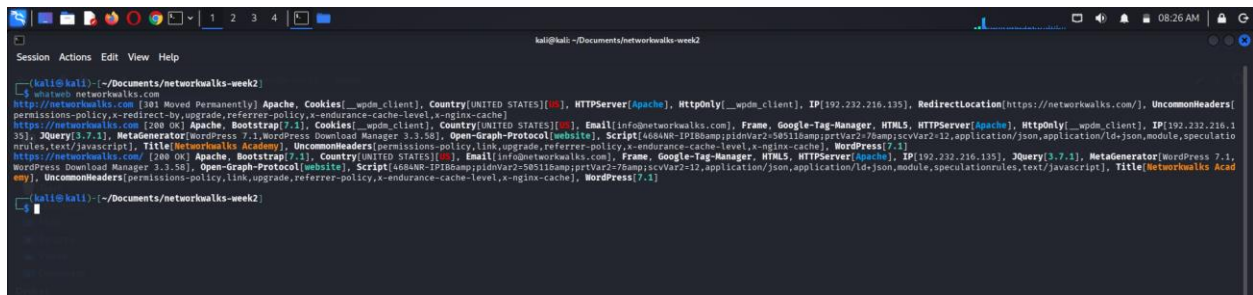
5.2 Web Technology Identification with WhatWeb

WhatWeb was used to identify technologies exposed by the target website.

Command

whatweb networkwalks.com

Result



```
kali@kali: ~/Documents/networkwalks-week2
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][0], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by:upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[_wpdm_client], Country[UNITED STATES][0], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[wordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pid=var2=58511&pr=var2=7&psc=var2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][0], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[wordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pid=var2=58511&pr=var2=7&psc=var2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

Observation

The results provided information about technologies and services that could be identified from the target's web application.

Examples of information that may be identified include:

- Web server technology
- JavaScript frameworks
- CMS platforms
- Web technologies
- Cookies
- HTTP headers
- Server-side technologies

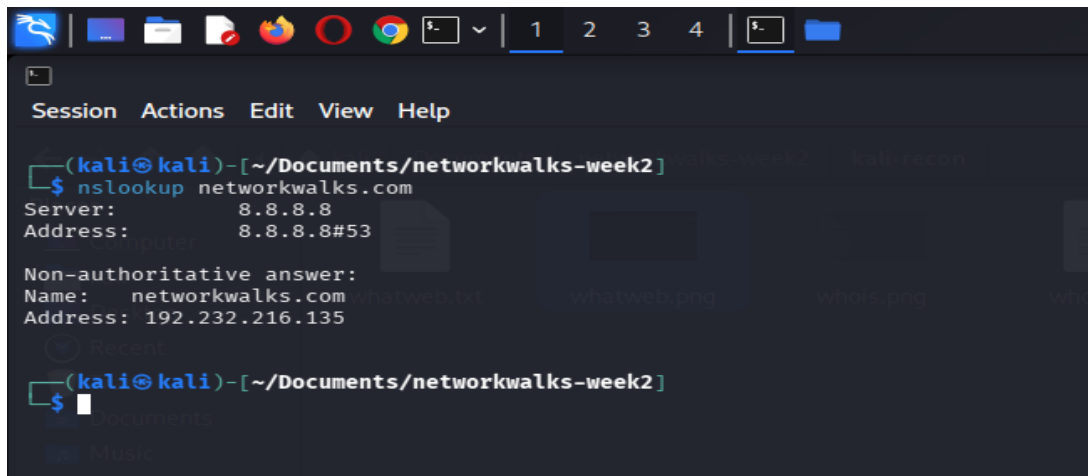
5.3 DNS Enumeration with NSLookup

NSLookup was used to resolve the target domain and inspect DNS records.

Commands

```
nslookup networkwalks.com
```

Result



```
(kali㉿kali)-[~/Documents/networkwalks-week2]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135

(kali㉿kali)-[~/Documents/networkwalks-week2]
$
```

Observation

DNS enumeration provided information about the target's domain resolution and DNS infrastructure.

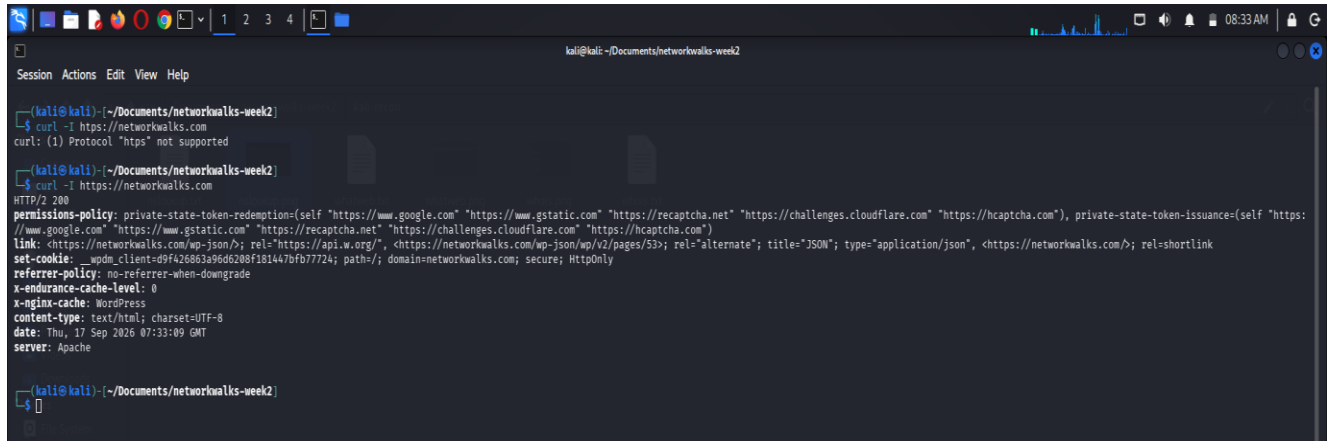
5.4 HTTP Header Analysis with cURL

cURL was used to inspect HTTP response headers from the target web server.

Command

```
curl -I https://networkwalks.com
```

Result



```
kali@kali:~/Documents/networkwalks-week2
$ curl -I https://networkwalks.com
curl: (1) Protocol "https" not supported

kali@kali:~/Documents/networkwalks-week2
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: wpdm_client=d9f426863a96d6208f181447bf77724; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Thu, 17 Sep 2026 07:33:09 GMT
server: Apache

kali@kali:~/Documents/networkwalks-week2
$
```

Information examined

- HTTP status code
- Server information
- Security headers
- Cache-related headers
- Redirect behavior
- Cookies and other response metadata

Observation

The HTTP headers provided additional information about how the target web application responds to HTTP requests.

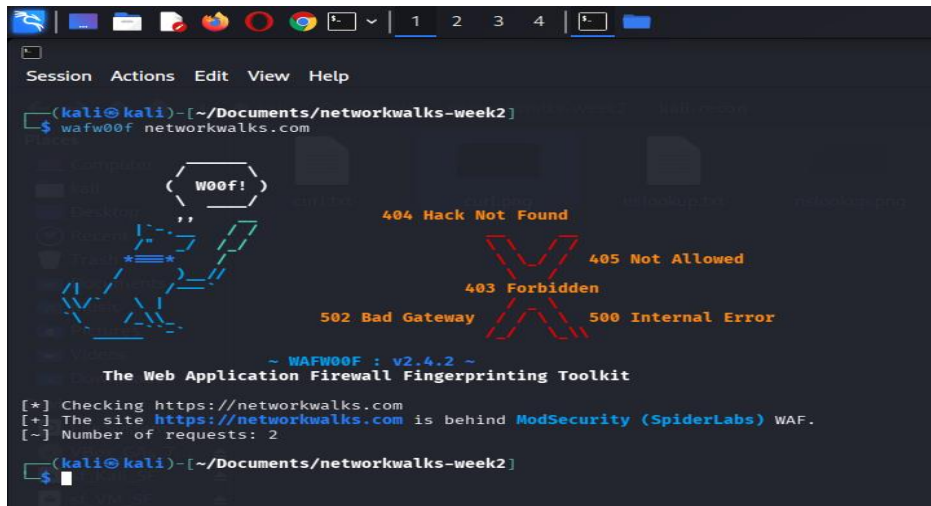
5.5 WAF Detection with WAFW00F

WAFW00F was used to determine whether a Web Application Firewall could be identified.

Command

wafw00f <https://networkwalks.com>

Result



```
(kali@kali) - [~/Documents/networkwalks-week2]
$ wafw00f networkwalks.com

W00f!

404 Hack Not Found
403 Forbidden
405 Not Allowed
502 Bad Gateway
500 Internal Error

~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[-] Number of requests: 2

(kali@kali) - [~/Documents/networkwalks-week2]
$
```

Observation

The result indicated:

WAF detected: YES

Identified WAF: ModSecurity (SpiderLabs) WAF

WAF detection results should be interpreted carefully because a WAF may be customized, hidden behind another service, or otherwise not identifiable by automated fingerprinting.

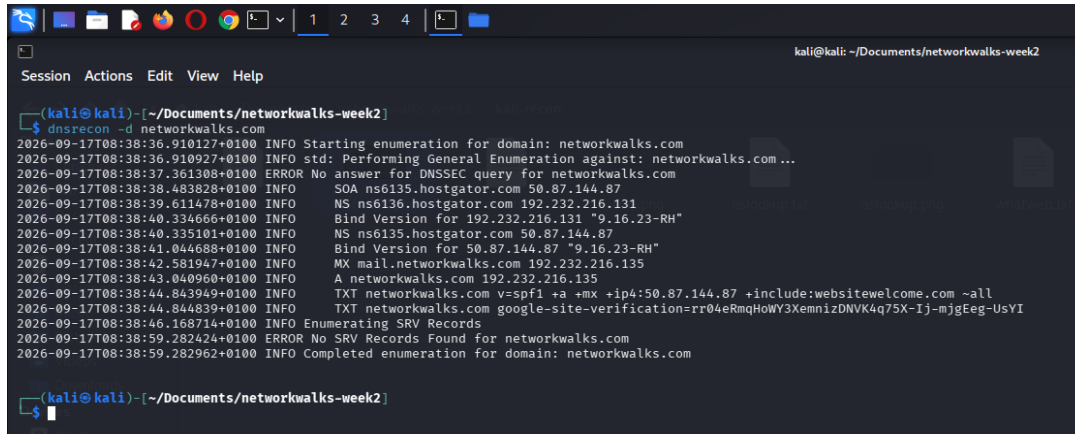
5.6 DNS Enumeration with DNSRecon

DNSRecon was used to perform additional DNS enumeration.

Command

```
dnsrecon -d networkwalks.com
```

Result



```
(kali@kali) - [~/Documents/networkwalks-week2]
$ dnsrecon -d networkwalks.com
2026-09-17T08:38:36.910127+0100 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T08:38:36.910927+0100 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-09-17T08:38:37.361308+0100 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-17T08:38:38.483828+0100 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-17T08:38:39.611478+0100 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-17T08:38:40.334666+0100 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-17T08:38:40.335101+0100 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-17T08:38:41.044688+0100 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-17T08:38:42.581947+0100 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-17T08:38:43.040960+0100 INFO A networkwalks.com 192.232.216.135
2026-09-17T08:38:44.843949+0100 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-17T08:38:44.844839+0100 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-09-17T08:38:46.168714+0100 INFO Enumerating SRV Records
2026-09-17T08:38:59.282424+0100 ERROR No SRV Records Found for networkwalks.com
2026-09-17T08:38:59.282962+0100 INFO Completed enumeration for domain: networkwalks.com

(kali@kali) - [~/Documents/networkwalks-week2]
$
```

Observation

DNSRecon provided additional information about DNS records associated with the authorized target.

The findings were reviewed to understand the externally visible DNS infrastructure without attempting unauthorized access.

6. Evidence Collected

The following evidence was collected during the exercise:

01-kali-footprinting/

└─ README.md

└─ Module1-Report-Recon-with-Kali

└─ docs/

└─ whois.txt

└─ whatweb.txt

└─ nslookup.txt

```
|— curl-headers.txt
|— wafw00f.txt
|— dnsrecon.txt
```

```
|— screenshots/
|   |— whois.png
|   |— whatweb.png
|   |— nslookup.png
|   |— curl-headers.png
|   |— wafw00f.png
|   └— dnsrecon.png
```

The raw output files contain the command results, while the screenshots provide visual evidence of the individual activities.

7. Key Learning Outcomes

Through this exercise, I gained practical experience with:

- Passive reconnaissance
- Domain enumeration
- DNS reconnaissance
- Web technology fingerprinting
- HTTP header analysis
- WAF identification
- Security information gathering
- Reconnaissance documentation

The exercise demonstrated how seemingly small pieces of publicly available information can be combined to build an understanding of a target's externally visible infrastructure.

8. Security and Ethical Considerations

Reconnaissance can reveal sensitive information about organizations and their infrastructure. For this reason, all activities were conducted within an authorized security-testing scope.

No attempt was made to gain unauthorized access, disrupt services, bypass security controls, or access private information.

Findings that could expose sensitive information should be appropriately redacted before being published in a public repository.

9. Disclaimer

I obtained permission from the relevant target/network owners before carrying out reconnaissance activities against the target networks used in this exercise.

All footprinting activities documented in this report were conducted for authorized cybersecurity training, assessment, and educational purposes.

The techniques demonstrated in this project should only be used against systems, networks, domains, and applications for which the tester has explicit authorization.

Unauthorized reconnaissance or security testing may violate organizational policies and applicable laws.

10. Conclusion

The Footprinting with Kali Tools exercise provided practical experience with the reconnaissance phase of a cybersecurity assessment.

By combining WHOIS, WhatWeb, NSLookup, cURL, WAFW00F, and DNSRecon, I was able to examine different aspects of an authorized target's externally visible infrastructure.

This exercise forms a foundation for subsequent activities involving enumeration, vulnerability assessment, and security testing within authorized environments.